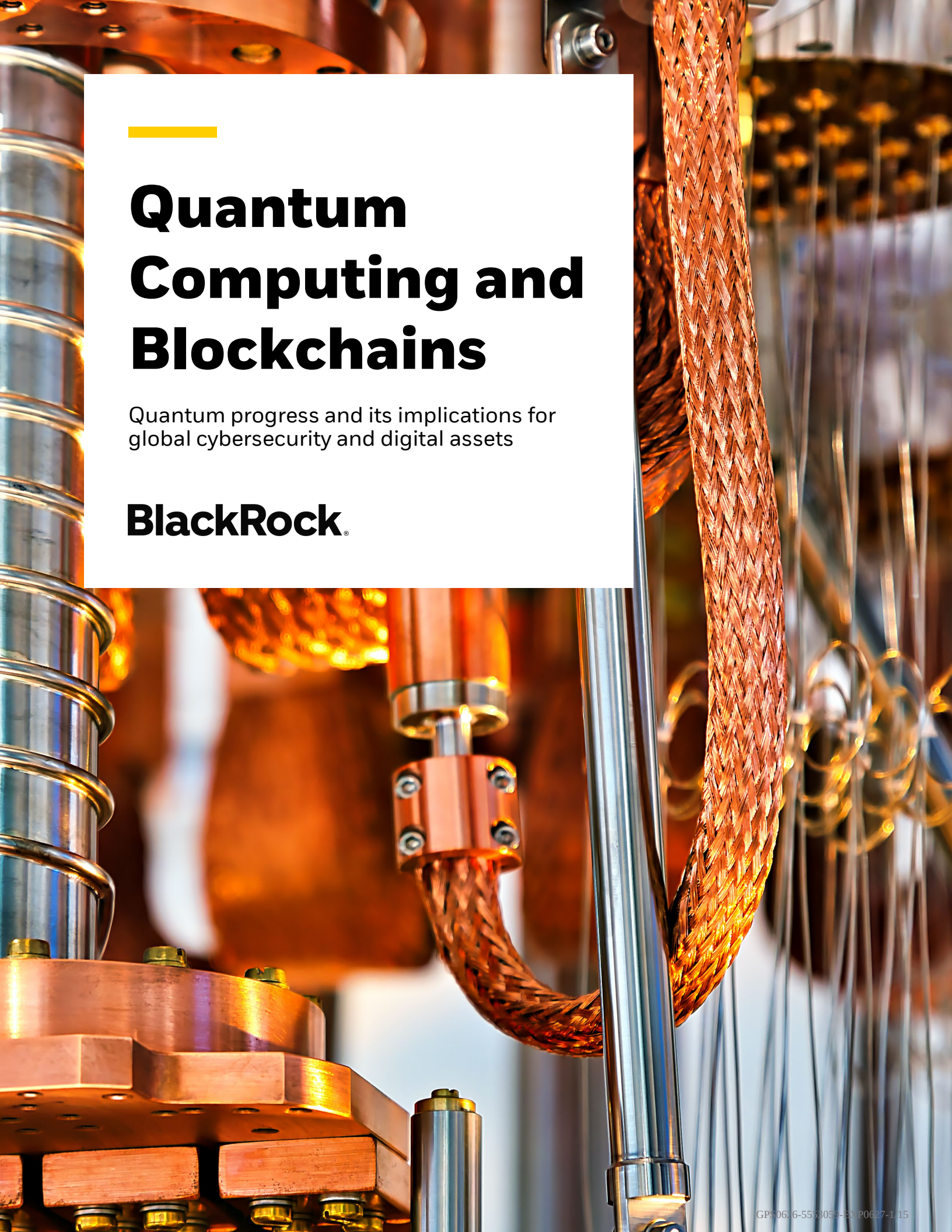


Quantum Computing and Blockchains

Quantum progress and its implications for global cybersecurity and digital assets

BlackRock®

GPS0626-5538059-BD P0627-1115

Executive Summary

Quantum computing has been the subject of growing attention in recent years, particularly due to its implications for blockchains and many other elements of modern cyber infrastructure. In this paper, we examine recent developments around quantum computing and its impact on the future of cybersecurity. While the potential implications of quantum computing across many domains of global cybersecurity are significant, this paper focuses on the intersection of quantum computing with blockchains. **In our view, quantum computing is likely to be a manageable risk for blockchains, subject to the industry's ability to upgrade swiftly and proactively to post-quantum cryptography in the coming years.**

Nearly half a century after Quantum Computing's conception in the early 1980s, the field is making progress on its journey from theory to reality, driven by ongoing hardware scaling and algorithmic innovation. In the future, functional quantum computers could open exciting new frontiers in biochemistry, materials science, and logistics optimization. At the same time, they also have the potential to eventually threaten the cryptography underlying most modern-day cybersecurity infrastructure, including bank accounts, emails, web browsers, SWIFT transfers, and blockchains. The emergence of a real-world Cryptographically Relevant Quantum Computer (CRQC), often referred to as "Q-Day", has shifted from a distant possibility many years or decades away to something that could arrive well within the coming decade. No functional CRQC exists today, and significant physics, technical, and engineering hurdles remain before one could become feasible. Yet recent breakthroughs have pulled forward quantum timelines from technology leaders in the field: Google is moving up its migration deadline to 2029 and IBM is targeting large-scale fault-tolerant quantum computing by 2029-2033 – bracketing the window in which quantum threats could become actionable.

The most widely used public-key cryptographic protocols in present-day cybersecurity, RSA and Elliptic Curve Cryptography (ECC), are underpinned by mathematical asymmetry: legitimate digital signatures are easy to verify, but key recovery attacks require solving problems that are infeasibly difficult even for the most advanced classical computers. This enables secure exchanges between parties over public channels, while deterring malicious actors from decrypting secure data. Quantum computers, on the other hand, can leverage quantum mechanical properties in combination with Shor's Algorithm (developed by Peter Shor in 1994) to solve these problems in a vastly more efficient manner.

Cryptographers and security researchers have long recognized emerging security risks posed by future quantum computers. Perceiving the quantum threat early, the National Institute of Standards and Technology (NIST) launched its Post-Quantum Cryptography (PQC) Standardization initiative in 2016, scrutinizing dozens of encryption candidates against both classical and quantum cryptoanalysis through a multi-year, global competition. This culminated in the selection of the first three lattice- and hash-based federal post-quantum standards in 2024, with additional evaluations underway. Concurrently, government agencies and large technology organizations worldwide have refined their post-quantum transition roadmaps and target migration timelines as more credible PQC standards emerge.

Authors

Will Su
Head of Digital Assets Research, BlackRock

Inish Crisson
Senior Software Engineer, Aladdin Digital Assets Lab

Robert Mitchnick
Head of Digital Assets, BlackRock

GPS0626-5553059-EXP0627-2/15

Cryptocurrencies are often one of the first topics discussed in the context of future risks from quantum computing, in large part because they represent the most readily monetizable "honey pot" for an attacker with a future CRQC to target. Evident in the name, Cryptocurrencies rely on cryptography to secure their blockchains, record transactions, and prove ownership. In the case of Bitcoin, the cryptographic hash function used to secure *the core blockchain transaction history* is called SHA-256, which is generally considered to be substantially resistant to quantum computing, with Grover's Algorithm (developed by Lov Grover in 1996) offering only a modest speed advantage for a quantum computer over classical computing. However, the cryptography used to secure *individual transactions* on Bitcoin and most other major blockchains, including Ethereum, currently employs digital signature schemes secured by Elliptic Curve Cryptography, making them likely quantum-vulnerable at some point in the future.

In our view, PQ migration for cryptocurrencies is eminently addressable from a technical standpoint, and the key challenge is one of timely coordination and implementation. When Bitcoin was first launched in the late 2000s, its pseudonymous creator Satoshi Nakamoto addressed the possibility of underlying cryptographic primitives being broken and discussed how the network could lock in the honest chain and transition to stronger algorithms. Nearly 2 decades later, given the risk that quantum computing poses to elliptic curve cryptography, this vision has moved closer to reality. The end-to-end process to build consensus around PQC protocols and timing, implement upgrades on the blockchain, and perform orderly migrations across the ecosystem will likely be a multi-year endeavor.

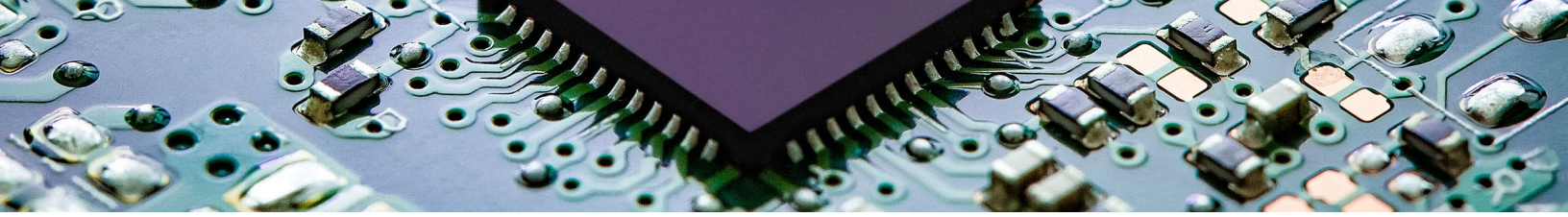
Different blockchains face varying challenges of technical difficulty and development coordination for PQ migration. For Bitcoin, upgrades are technically straightforward, but development is relatively decentralized and there is no current consensus on PQ encryption/signature schemes, migration timelines, and the optimal specific implementation mechanisms. Ethereum, on the other hand, has a detailed migration path guided by the Ethereum Foundation, but faces greater technical challenges given its multi-layered architecture, its Proof-of-Stake consensus mechanism, and the complexity of its smart contract architecture. In concert with the underlying blockchain protocols, the broader digital assets ecosystem, including exchanges, custodians, and validators, will also need to upgrade hardware, software, and policies to become quantum-secure over a multi-year timeframe.





Ultimately, we view implementation PQC signature schemes in Bitcoin as a much lower hurdle compared to the many remaining technical challenges towards a functional CRQC, recent algorithm progress not withstanding. In sum, quantum computing presents a moment of risk and opportunity for digital assets. We believe executing successful post-quantum migrations will lead to stronger digital infrastructure, greater blockchain adoption, and likely higher cryptocurrency valuations if this threat becomes de-risked and eventually resolved, in our view. The task before the industry is to execute on an orderly migration comfortably in advance of Q-Day while minimizing disruptions to blockchain functionality and mitigating potential impacts to market structure.

GPS0626-5553059-EXP0627-3/15



Elliptic Curve Cryptography: the Mathematical Foundation of Modern Cybersecurity

Modern-day secure communications, including the internet, financial networks, and blockchains, leverage cryptography to transmit sensitive information over insecure, public domains. The two most used asymmetric cryptographic protocols are RSA (Rivest-Shamir-Adleman) and ECC (Elliptic Curve Cryptography), which utilize public-private key pairs produced by large prime multiplication and point addition respectively. Even though the public key (or a hash of it) is visible to everyone and used to receive messages and transactions, the associated private key remains a secret because deriving it is computationally impossible for even the most advanced classical supercomputers.

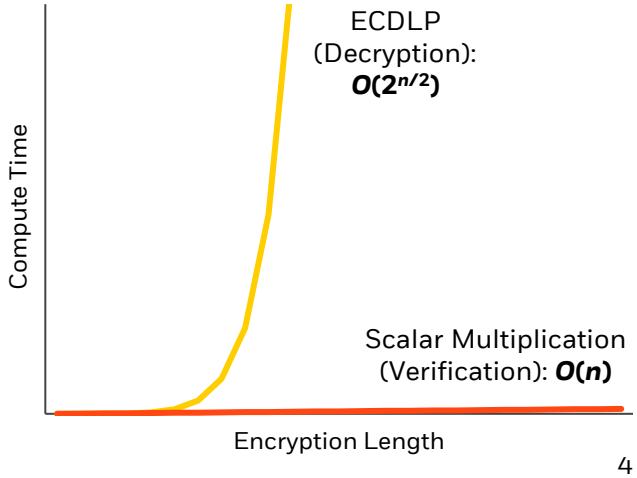
Bitcoin, and most cryptocurrency blockchains, utilize ECC to facilitate the secure movement of bitcoin between individual addresses on the blockchain. The Bitcoin blockchain is simply a distributed ledger that tracks ownership of value through unspent transaction outputs, or UTXOs, which are discrete quantities of bitcoin created and consumed via publicly visible transactions. UTXOs are locked to a public key identifier unique to each address. While any address can receive UTXOs via its public key, a hidden, corresponding private key is needed to "unlock" or send its UTXO balance.

The private key k for each address is a large, randomly generated integer between 1 and 2^{256}, and the public key, $k \cdot G$, is generated by performing point additions starting from a fixed point, G, along the elliptic curve.[1] Naively, this would require up to 2^{256} operations – approaching the number of atoms in the observable universe and clearly infeasible to calculate, even for the fastest classical supercomputers today. However, an algorithm known as "double-and-add" reduces this to just a few hundred operations on average by processing the key one bit at a time.

To "spend" Bitcoin, i.e., to transfer the UTXO balance held in one address out to another, the owner broadcasts a digital signature to the blockchain. The signature weaves the private key, without revealing it, with a hashed message (for example, how much BTC to send to another address). Validators on the blockchain then verify the signature is legitimate by performing a series of calculations involving scalar multiplications (still in linear time) and the public key, which confirms that the signature could only have been created with knowledge of the corresponding private key.[2]

In this way, ECC and RSA encryptions are elegant "trap doors": it is easy to prove someone knows the private key based on digital signatures, but functionally impossible to reverse-engineer what the private key is with current computing methods. Signing and verification compute in linear time, $O(n)$; private key recovery is an elliptic curve discrete logarithm problem (ECDLP) with no shortcuts and computes in exponential time, $O(2^{n/2})$. Bitcoin and Ethereum's current 256-bit ECC cryptography would take the fastest classical supercomputers today millions to billions of years to break – in other words, effectively impossible. The same is true for other systems that use ECC and RSA (e.g., HTTPS, VPNs, SWIFT interbank transfers, and government communications).

Figure 1: Illustrative Classical Compute Time, ECC Signing and Verification vs. Decryption



GPS0626-5553059-EXP0627-4/15



Quantum Computing, if Scaled, Could Threaten Current Encryption Protocols

"If you think you understand quantum mechanics, you don't understand quantum mechanics."

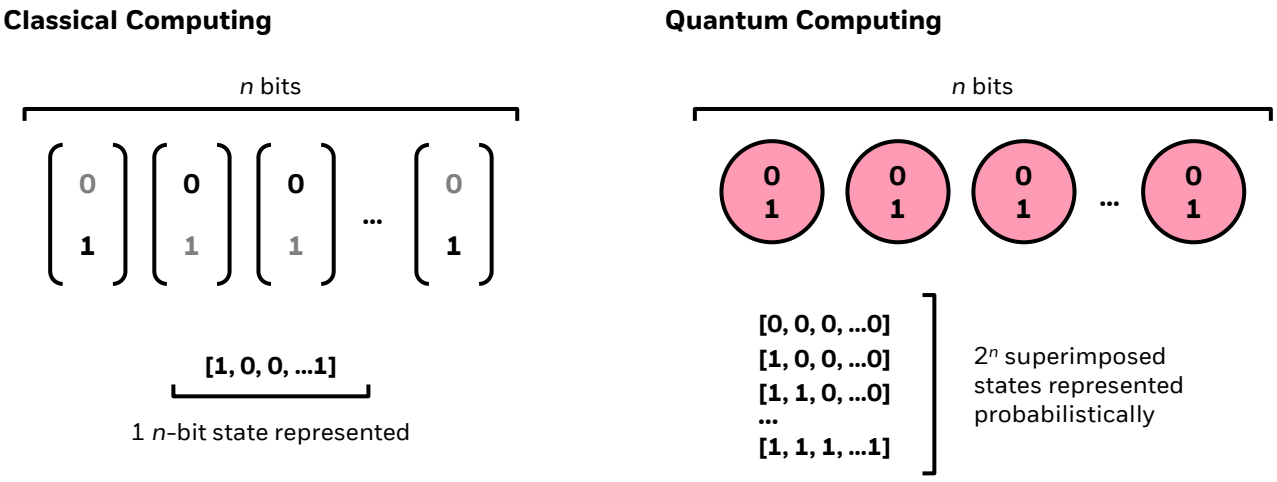
– Richard Feynman

The foundations of modern-day cryptography become challenged in the quantum world. This is not because quantum computers (QCs) run faster. Rather, QCs are particularly efficient at teasing out hidden patterns in large datasets by leveraging unique properties of quantum physics and employing quantum algorithms to solve classically infeasible problems like ECDLPs in as little as days to minutes. This could allow malicious actors to extract private keys from public information, compromise the ownership of digital assets, and corrupt the public blockchain ledger.

To be clear, no functional CRQC exists today, and significant physics, technical, and engineering hurdles remain. But recent breakthroughs in QC technologies, while not an immediate threat, introduce greater urgency to refine and execute post-quantum (PQ) migrations. For blockchains, an orderly transition likely entails a multi-year migration away from ECC and towards quantum-resilient protocols like hash- and lattice-based cryptography that have no known weaknesses for quantum algorithms to exploit. It also requires deep coordination within blockchain communities.

But first, it is helpful to understand how quantum computers work on a surface level. In classical computing, bits represent information in discrete, "either-or" 0 and 1 states. In quantum computing, quantum bits known as qubits can simultaneously represent both 0 *and* 1 states probabilistically under quantum **superposition**, and additional information is encoded in correlations between qubits under quantum **entanglement**.[3] Quantum states can be physically represented in several ways, including electron rotation (spin up/spin down), photon polarization (horizontal/vertical), or current flow in a superconducting circuit (clockwise/counterclockwise). While *n* classical bits can only represent one *n*-bit state at any given time, *n* qubits can represent 2^n superimposed quantum states simultaneously. A QC with ~300 logical, error-corrected, and entangled qubits can superimpose more states than there are atoms in the universe.

Figure 2: Qubits Can Simultaneously Represent Many States Under Superposition

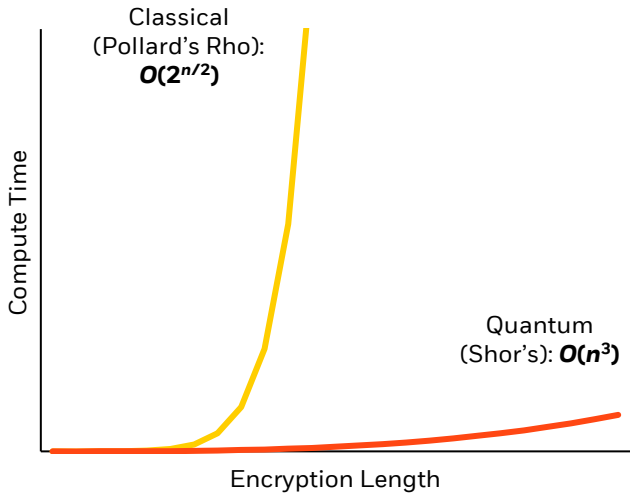


GPS0626-5553059-EXP0627-5/15

To “break” elliptic curve cryptography, a hypothetical CRQC deploys **Shor’s Algorithm**, which transforms the classical, brute-force ECDLP into a period-finding problem that can be solved significantly faster[4]. This is possible because elliptic curve equations have hidden patterns, or periodicities, related to the private key *k*. To oversimplify, the quantum computer simultaneously calculates all possible iterations along the elliptic curve given a public key. The resulting states are held in a quantum superposition and contain a pattern related to the private key that is too weak to be measured directly. This “position” state is transformed into its “frequency” state via a Quantum Fourier Transform (QFT), where noise is reduced and the target pattern is amplified[5]. The periodicity of this pattern is then measured and used to solve for the private key.

The transformed period-finding problem runs in polynomial compute time, $O(n^3)$. This is meaningfully more feasible than the classical, exponential compute time, $O(2^{n/2})$, for $n = 256$, the bit size of ECC keys currently used in Bitcoin and Ethereum. In short, this extreme speed-up in key recovery is the existential threat CRQCs pose to the foundational cryptography widely employed today, in technologies spanning from cryptocurrencies to traditional finance and most telecommunications channels.

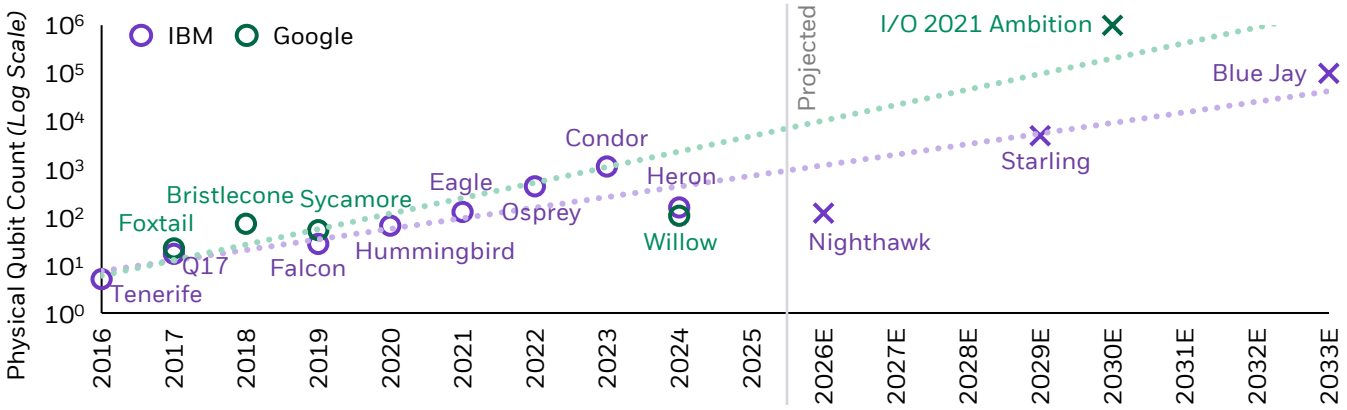
Figure 3: Illustrative Classical vs. Quantum Compute Time, ECC Decryption



The Timeline to Q-Day Has Likely Shortened Following Recent Algorithmic Breakthroughs

Quantum computing is advancing today across multiple architectures, designs, and labs, with varying degrees of technical maturity and proven functionality. For example, IBM and Google have been developing Superconducting Circuits, the most commercially deployed (>40% of QPUs)[6] quantum architecture, for more than a decade, while emerging architectures like Photonic and Neutral Atoms are expected to grow faster and gain share in coming years. IBM and Google’s development history and quantum roadmaps,[7,8] as measured by physical qubit count, have roughly tracked Moore’s Law-like exponential growth.

Figure 4: IBM & Google Quantum Roadmap (Superconducting Circuits)[7,8]



IBM: Target Blue Jay processor with 100,000 physical qubits by 2033+. Google: Target 1 million physical qubits “within the decade” at I/O 2021.
Source: IBM Quantum Computing, Google Quantum AI.

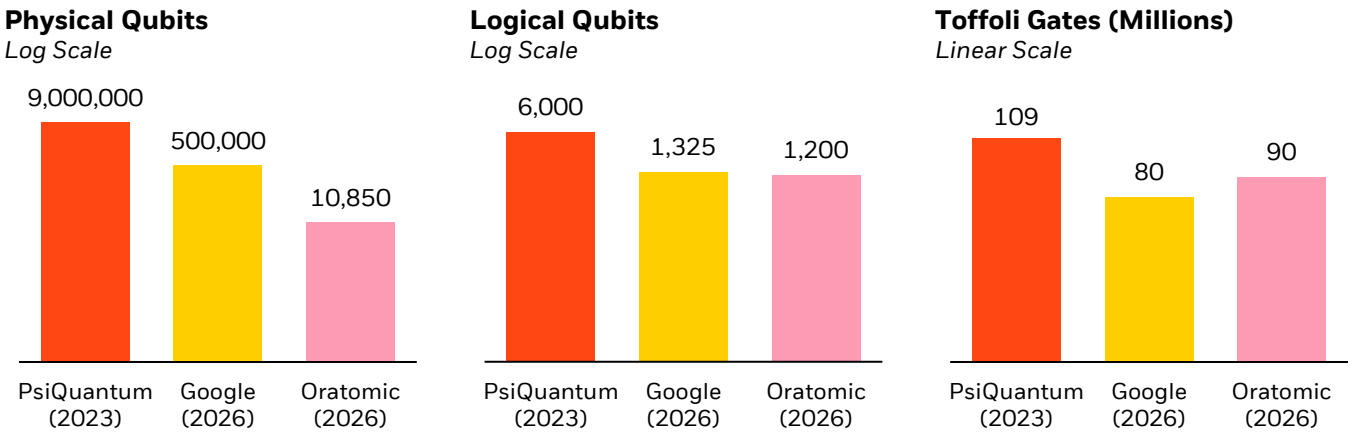
GPS0626-5553059-EXP0627-6/15

While the growth in physical qubit count is an important headline measure of quantum progress, it is a misleading figure in isolation. Quantum information is extremely fragile, and physical qubits can quickly decohere and drop out of quantum states; as a result, inherent error rates are too high for practical use in isolation, and multiple physical qubits must be combined through quantum error correction to produce a single usable, fault-tolerant logical qubit. This ratio, known as "overhead", has an outsized impact on the timeline of CRQC development, and has been the focus of recent QC breakthroughs. For example, after unveiling the Condor quantum processing unit (QPU) in 2023 with 1,121 physical qubits, IBM followed with Heron (2023) and Nighthawk (2025) QPUs featuring 156 and 120 physical qubits – a large stepdown in volume.

Performance, on the other hand, improved materially: Heron and Nighthawk delivered significant improvements in both error reduction and physical efficiency. In 2024, Google's Willow QPU demonstrated below-threshold error correction for the first time, with each step-up in surface code size – from grids of 3x3 encoded qubits to 5x5 and to 7x7 – reducing error rate by half[9]. This achievement opens a more credible path to scale up the compute power necessary for larger-scale quantum operations.

Additional leaps have also been made in recent years in terms of algorithm optimizations that reduce physical resource estimates for breaking RSA and ECC cryptography. Google's March 2026 whitepaper optimized circuit compilation specifically on solving ECDLP-256[11]. Along with overhead reduction, this reduced the physical CRQC resource estimate dramatically, with a 4-5x reduction in logical qubits and a 18x reduction in physical qubits compared with previous benchmark estimates from PsiQuantum[12]. Also in March 2026, researchers from Caltech/Oratomic released a whitepaper highlighting breakthroughs in error correction tailored to their neutral atom architecture[13]. The paper estimated that as few as 10,000 reconfigurable physical qubits are needed to break ECC-256 cryptography (note that neutral atom is a less mature design from an engineering perspective, and not directly comparable to resource estimates for superconducting circuits).

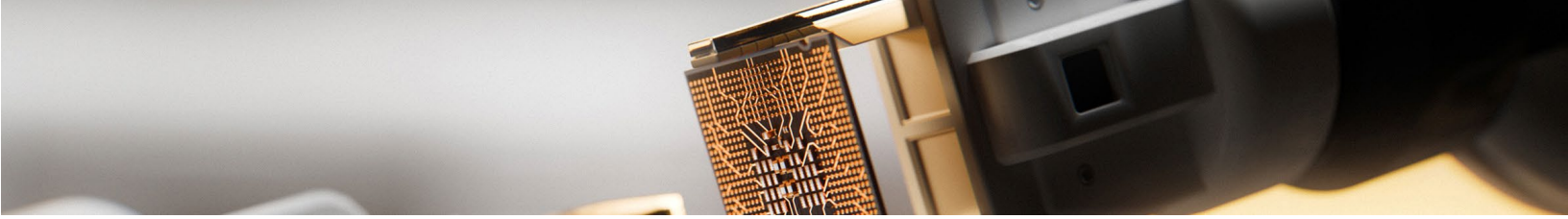
Figure 5: Estimated Quantum Resource Needed for ECC-256 Decryption



Source: Google Quantum AI, CalTech/Oratomic, PsiQuantum.

As physical engineering continues to grow and as software, error correction, and algorithms become further optimized, estimates for Q-Day by leading academics, experts, and industry practitioners have consequently been brought forward in recent years. That said, many engineering challenges remain in the quantum journey. Quantum computing generally demands extreme environmental isolation over much longer periods than demonstrated, and a functional CRQC demands orders of magnitude improvements in physical qubit count and error reduction from state-of-the-art systems today.

Fundamentally, it is a much less daunting task to upgrade current cryptographic systems (including Bitcoin) to a quantum-secure standard than it is to build a CRQC, from where quantum computing progress stands today. Thus, advantage remains decidedly with the Defense at the current juncture, as we will explore in the next section.



Post-Quantum Cryptography (PQC) Standards Are Already in Use as Post-Quantum Transition Roadmaps Pull Forward

In our view, post-quantum migration is a manageable challenge from a technical perspective. Cryptographers and security researchers have spent years developing, battle-testing, and refining digital signature and verification schemes that are quantum-resistant, i.e., they cannot be decrypted by either classical or quantum methods in feasible compute time.

The National Institute of Standards and Technology (NIST), an agency under the Department of Commerce, has led the drive towards quantum-resistant standards. Having recognized early that QCs would eventually break many mainstream cryptographic algorithms, including RSA and ECC, NIST launched a global competition in 2016 for open submissions of post-quantum public-key cryptographic algorithms. After multiple rounds of evaluation and open testing (in which two of seven finalists were broken), three post-quantum standards have been finalized as of today: one for Key Encapsulation (**ML-KEM**) and two for Digital Signatures (**ML-DSA** and **SLH-DSA**).

Additional, specialized protocols are being further evaluated for final standardization, including FN-DSA (smaller lattice-based signatures) and HQC (error correcting codes-based key encapsulation). Because both primary key encapsulation and digital signatures standards are lattice-based, NIST is evaluating additional submissions to diversify this post-quantum portfolio, with further standards targeted towards the end of the decade.

Figure 6: Current NIST Post-Quantum Cryptography Standards

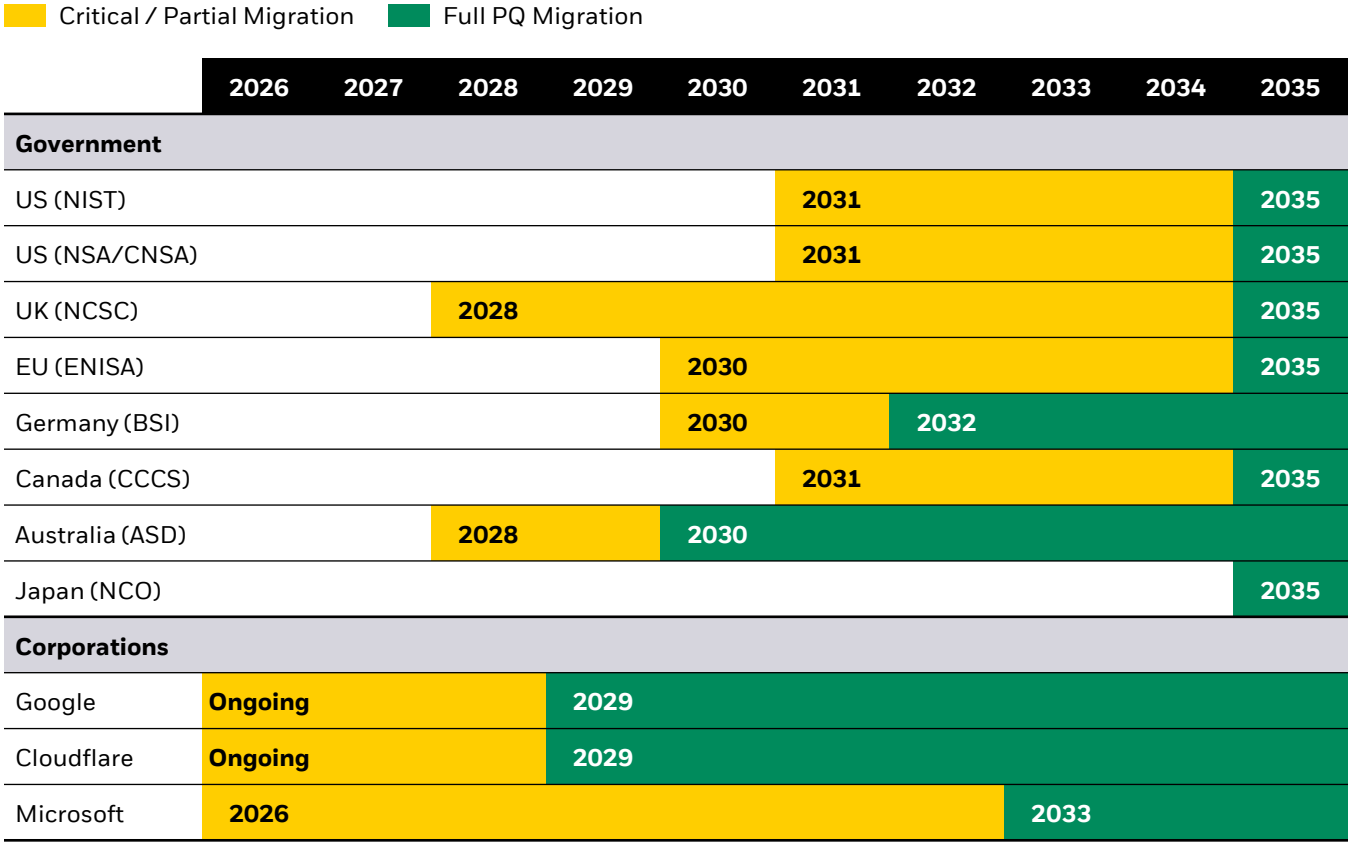
Standard	Name	Previous Name	Application	Position	Algorithm
Finalized Standards					
FIPS 203	ML-KEM	CRYSTALS-Kyber	Key Encapsulation	Primary	Lattice-based
FIPS 204	ML-DSA	CRYSTALS-Dilithium	Digital Signatures	Primary	Lattice-based
FIPS 205	SLH-DSA	SPHINCS+	Digital Signatures	Backup	Hash-based
Progressing to Finalize					
FIPS 206	FN-DSA	FALCON	Digital Signatures	Specialized	Lattice-based
N/A	HQC	Hamming Quasi-Cyclic	Key Encapsulation	Backup	Codes-based

Source: National Institute of Standards and Technology.

A sampling of announced post-quantum migration timelines (Figure 7) shows that governments globally have largely converged with NIST's 2035 deadline[14] for a full migration away from quantum-vulnerable protocols, although timelines differ for the implementation date for critical and national security-sensitive applications. Notably, NIST's primary selections, ML-KEM and ML-DSA, feature prominently across most PQ cryptography schemes. For private companies, both Google[15] and Cloudflare[16] pulled forward PQ migration deadlines to 2029, citing the March 2026 Google Quantum AI whitepaper as an impetus for the acceleration. We expect more public and private entities will follow and accelerate their own migration deadlines. The urgency of PQ migration is arguably greater for governments and technology companies than for blockchains because of "harvest now, decrypt later" attacks: adversaries can stockpile encrypted information that remains sensitive for decades – including state secrets, personal records, and corporate data – and decrypt them once a CRQC becomes available.

GPS0626-5553059-EXP0627-8/15

Figure 7: Overview of Announced Public / Private Post-Quantum Migration Timelines[17-24]



Source: Government agency reports (NIST, NSA/CNSA, NCSC, ENISA, BSI, CCCS, ASD, NCO), company reports (Google, Cloudflare, Microsoft).

PQ Migration for Blockchains: Roadmaps Are Forming, Coordination and Execution Are Key Milestones

Blockchain communities have not stood still in the face of growing quantum threats: Bitcoin, Ethereum, and other blockchains have made varying progress towards post-quantum readiness. The radical transparency of blockchains offer some advantages: the potential attack surface is well-understood and quantifiable, and the open, adversarial review process by economically motivated participants should result in robust and well-tested implementations once activated. Compared to sweeping upgrades required across global corporate and government cyber infrastructure, the scope of blockchain upgrades is considerably narrower: for Bitcoin, it centers on replacing a single digital signature algorithm. That said, decentralized blockchains also face unique challenges in managing PQ migrations relative to traditional, centralized cyber infrastructure: without a singular authority enforcing key decisions like PQC standards, timelines, and implementation processes, blockchain communities must build social consensus and coordinate implementation among many stakeholders. Larger PQ signatures also pose technical challenges such as increased pressure on block space, or impacts to attributes such as transaction speed, privacy, and application flexibility.

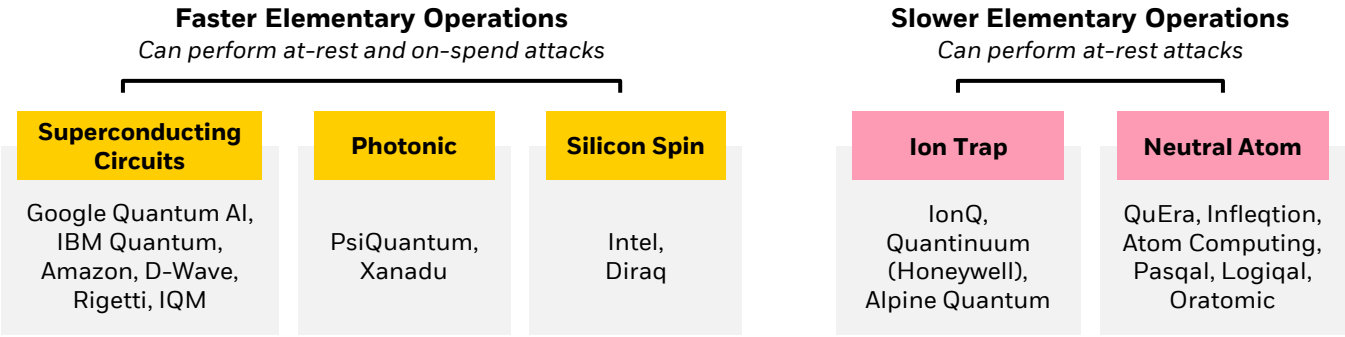
GPS0626-5553059-EXP0627-9/15

Importantly, the Proof of Work mechanism that secures the integrity of Bitcoin's blockchain utilizes the hash function SHA-256, which is largely considered quantum-resistant.

Grover's algorithm provides only a quadratic speedup, an advantage which would be absorbed by the network's automatic difficulty adjustment rather than undermining consensus. The primary attack surface focuses on the security and integrity of ownership via addresses and wallets.

There are two ways a CRQC could attack cryptocurrencies and authorize illegitimate transactions: Slow (At-Rest, Long-Range) and Fast (On-Spend, Short-Range), likely depending on which quantum architecture reaches CRQC breakout first. Architectures including Superconducting Circuits, Photonic, and Silicon Spin utilize lighter electrons and photons to perform quantum calculations faster, while Ion Trap and Neutral Atom designs use atomic-mass objects that compute at slower frequencies.

Figure 8: QC Architecture Groups by Elementary Operations Speed

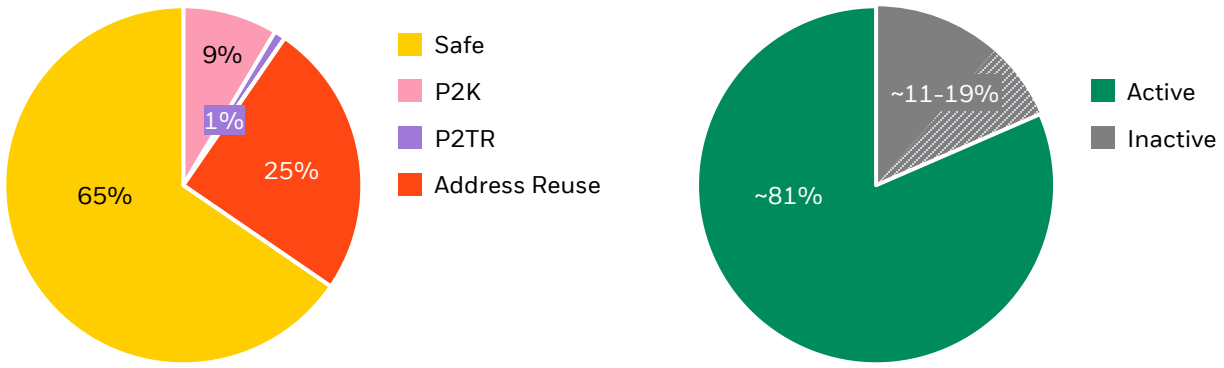


Source: Company reports, filings, and whitepapers.

An address is vulnerable to long-range attacks when its public key has been exposed on the blockchain, either through the address type it employs or via re-use of an address. This makes such addresses a target for both fast-clock and slow-clock quantum computers. If a fast-clock CRQC emerges and can recover the private key faster than the block time (i.e., the time needed to validate legitimate transactions), then short-range attacks could also threaten addresses without an already-exposed public key whenever they initiate a transaction and reveal the unhashed public key in the mempool.

Nearly 7 million, or ~35%, of Bitcoin circulating supply are potentially vulnerable to long-range attacks[25]. This includes 1.9 million BTC held in addresses with script types that expose unhashed public keys (P2PK, P2TR, and P2MS), and an additional 5 million held in addresses with script types that hash public keys but have exposed them in past transactions and still hold UTXOs. The majority of Bitcoin is held in addresses with script types (P2PKH, P2SH, P2WPKH) that scramble public keys twice using hash functions, mitigating the threat vector from long-range attacks.

Figure 9: Current Bitcoin Circulating Supply Vulnerable to Long-Range Attacks



Source: Project Eleven, Chainalysis.

We believe UTXOs held in quantum-vulnerable addresses will swiftly migrate once PQ protocols are activated on the blockchain, driven by overwhelming financial risks of inaction. However, migration is not possible for UTXOs held in inactive addresses with deceased owners or lost private keys, including the ~1.1 million in P2PK addresses widely believed to be owned by Satoshi Nakamoto. Chainalysis estimates 2.3–3.7 million BTC (11–19% of circulating supply) may be permanently lost based on long-term dormancy analysis[26]. The treatment of these holdings in the PQ migration remains an active topic of debate within the bitcoin community today.

Short-range attacks represent a theoretical future risk to all ECC-based addresses when they initiate transactions, regardless of script type. Short-range attacks take advantage of the short window when legitimate transactions are broadcast and unhashed public keys become visible in the mempool, waiting for the next block to be mined. Historically, short-range attacks were considered as significantly less practical than long-range attacks, as they would require a CRQC capable of breaking ECC within Bitcoin's ~10-minutes average block time – a far higher performance bar than the days, hours, or weeks assumed for long-range attacks. However, Google Quantum AI researchers estimated in their March 2026 whitepaper that a theoretical fast-clock CRQC would put short-range attacks in play alongside long-range attacks. If realized, this would allow an attacker to obtain the private key via Shor's and broadcast a fraudulent transaction, before the original, genuine transaction is confirmed.

Bitcoin proposals are historically infrequent, take years to implement, and have been at times contentious. This is a deliberate part of Bitcoin's decentralized, intentionally rigid, architecture. To propose an upgrade, a developer submits a Bitcoin Improvement Proposal (BIP) to a mailing list of key developers, who debate and critique the proposal via channels such as GitHub and Google Groups. BIPs that survive an extensive review are formalized and shipped over Bitcoin Core software updates and formally implemented once the network reaches sufficient signaling thresholds.

Today, several BIPs related to PQ migration have been accepted into the repository[27], although they remain in the draft stage, with additional proposals waiting to be assigned BIP numbers. While further progress needs to be made, early contours of agreement are emerging.

Broadly, we expect Bitcoin's ultimate, finalized migration roadmap to address several key issues over coming years:

- **Adopt a PQ signature scheme**: This will likely involve adding an option to the Bitcoin protocol to support new quantum-resilient signature mechanisms such as Lattice- or Hash-based signatures. Signature size is an area of active development: SLH-DSA, for example, uses ~7.8 KB signatures, >100 times larger than ECC/Schnorr signatures used today. More compact PQC schemes, such as SHRINCS/SHRIMPS[28], are being designed to be more compatible with bitcoin's limited block capacity, but remain in development.

- **Leverage existing mechanisms like Taproot's script architecture to accelerate migration timeline**: Taproot's script-path spending mechanism provides a foundation for quantum-resistant output designs. BIP-360 proposes removing the quantum vulnerable key path while preserving script-path functionality, giving users immediate protection against long-exposure attacks. Activating PQ opcodes via tapscript would then extend protection to short-range attacks.

- **Implement UTXO Output and Signature Migration policies:** The treatment of "lost" UTXOs is a contentious and complex topic, with debate lines drawn between Do Nothing/Ignore (keeping the unmigrated and vulnerable UTXOs spendable on the blockchain) and Deprecate (render these balances unspendable after a certain pre-determined point in time). In our view, this debate will likely resolve most constructively once a PQ signature scheme has already been adopted and is active on the Bitcoin network. We also note that should deprecation occur, a series of recent technical proposals create tangible pathways for rightful owners to reclaim UTXOs via zero-knowledge proofs, even after PQ migration is complete.

- **Evaluate strategies to accommodate larger, quantum-secure signatures longer-term:** For example, the network could adjust transaction weight calculations to avoid disproportionately penalizing PQ-secured transactions.

Ethereum's migration plan is well-defined after years of research and planning but involves significantly more technical complexity than Bitcoin's challenges. In early 2026, Vitalik Buterin, Ethereum's co-founder, delineated four quantum vulnerabilities across Ethereum's three layers[29]: (1) BLS signatures in the Consensus Layer, (2) KZG proofs used in the Data Layer, (3) EOA (Externally Owned Accounts) signatures, and (4) zero-knowledge proofs in the Application Layer. In simple terms: how validators vote, how data is efficiently verified, how users sign transactions, and how apps prove computations are all based on quantum-vulnerable elliptic curves, and need to be upgraded to PQ cryptographic alternatives.

The Ethereum Foundation concurrently released a "L1 Strawmap" – a draft long-term roadmap of seven upcoming network updates/hard forks between 2026 and 2029, of which five directly address quantum vulnerabilities[30]:

- **EOA Signatures - Forks H*/Hegota (2H2026) and J* (2027):** The Hegota update will ship EIP-8141 and add native account abstraction, which allows user accounts to adopt any signature algorithm. This opens the upgrade path to quantum-resistant signature schemes in the future, especially smaller, more efficient protocols currently being developed. J* will deploy PQ signature precompiles on the Execution Layer, which enables affordable on-chain verifications for smart contract accounts.

- **BLS validator signatures – Forks I* (2027) and L* (2028):** These updates allow validators, which secure the blockchain via proof-of-stake, to register post-quantum public keys alongside existing BLS keys and eventually provide attestations on the Consensus Layer using hash-based (leanXMSS) signatures instead of quantum-vulnerable BLS signatures. Because hash-based signatures are much larger and can't be natively combined like BLS signatures, SNARK-based aggregation (leanVM) will be introduced to compress many individual signatures into smaller proofs and keep consensus scalable on the network.

- **Data blob handling – Fork M* (2029):** Data blobs allow Layer 2 rollups to cheaply post transaction data to Ethereum for public verification without burdening the main Execution Layer. The integrity of this data is verified using KZG commitments, which are based on quantum-vulnerable elliptic curve pairings. This update, which requires further engineering work to finalize, will transition blob verification to quantum-resilient STARK-based proofs. It also provides protocol-level aggregation of PQ signatures.

Additional updates beyond 2029 focus on bringing the post-quantum transition to finality, fully phasing out legacy cryptography currently used in Consensus (BLS), transactions (ECC-based signatures), and data sampling (KZG commitments and proofs). Vulnerability (4), Application Layer zero-knowledge (ZK) proofs, will also need to be addressed longer-term: additional development work is needed to migrate from vulnerable SNARK verifications to quantum-safe but more expensive STARK verifications.

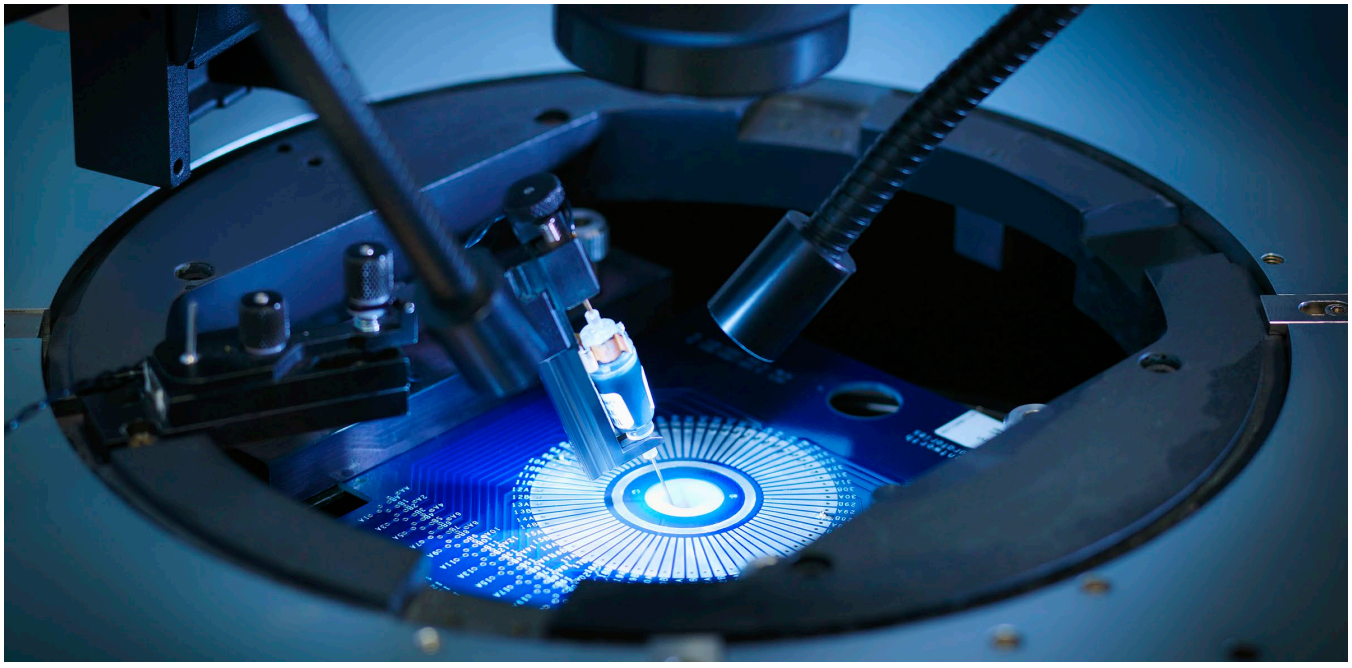
Broader ecosystem participants have started early-stage PQ preparations. Some blockchains have already initiated early post-quantum upgrades: for example, Solana has introduced a Winternitz Vault enabling users to migrate assets to PQ addresses[31], and Algorand has executed a PQ-signed transaction on their mainnet using post quantum cryptography[32]. Beyond the blockchains themselves, many other key players in the digital asset ecosystem are starting to mobilize towards post-quantum readiness, although current efforts remain largely exploratory. Many wallet providers, custodians, and exchanges have not yet disclosed concrete PQ migration plans, but we see a clear momentum shifting towards a greater focus on quantum readiness.

GPS0626-5553059-EXP0627-12/15

Conclusion: The Quantum Risk and Opportunity for the Future of Digital Assets

Global cybersecurity infrastructure stands at an important inflection point as quantum computing advances. Governments and large technology firms have finalized post-quantum cryptographic standards and accelerated their quantum-readiness timelines. Digital assets including Bitcoin and Ethereum are technically positioned for migration; a harder problem is coordinating timelines and rolling out upgrades across decentralized networks in an orderly manner. That said, **it is a much less daunting task to upgrade current cryptographic systems (including Bitcoin, Ethereum, and others) to a quantum-secure standard than it is to build a CRQC from where quantum computing progress stands today. Thus, advantage remains decidedly with the Defense, at the current juncture.**

Quantifying the drag of "quantum-negative" headlines on digital asset prices is difficult. We view the quantum risk as one of the few remaining walls of worry for the asset class and expect tangible progress on well-executed PQ migrations to be rewarded with higher valuations over time. Assuming a successful migration, we believe blockchains should emerge stronger and more resilient from this transition, supporting the long-term continued growth and proliferation of digital assets over the coming decades.



Endnotes

1. Lynn, "Elliptic Curves," https://crypto.stanford.edu/pbc/notes/elliptic/explicit.html.
2. Brandão, "Notes on Threshold ECDSA/Schnorr Signatures," https://csrc.nist.gov/csrc/media/Presentations/2022/crclub-2022-08-10/images-media/20220810-crypto-club-slides--threshold-EdDSA.pdf.
3. Preskill, "Quantum Computing in the NISQ era and beyond," https://quantum-journal.org/papers/q-2018-08-06-79/pdf/.
4. Zhu, Yue & Fan, "Shor's Algorithm and the Period Finding Problem," https://math.mit.edu/research/highschool/primes/materials/2019/December/Fan-Yue-Zhu_Shor_s_Algorithm_and_the_Period_Finding_Problem.pdf.
5. IBM Quantum Platform, "Quantum Fourier transform," https://quantum.cloud.ibm.com/learning/en/modules/computer-science/qft.
6. MIT Initiative on the Digital Economy, Quantum Index Report 2025, https://qir.mit.edu/wp-content/uploads/2025/06/MIT-QIR-2025.pdf.
7. IBM Quantum Roadmap, https://www.ibm.com/quantum/hardware.
8. Google Blog, "Unveiling our new Quantum AI campus," https://blog.google/innovation-and-ai/products/unveiling-our-new-quantum-ai-campus/.
9. Google Quantum AI computing roadmap, https://quantumai.google/roadmap.
10. Google Blog, "Meet Willow, our state-of-the-art quantum chip," https://blog.google/innovation-and-ai/technology/research/google-willow-quantum-chip/.
11. Babbush, Zalcman, Gidney et al, "Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities: Resource Estimates and Mitigations," https://quantumai.google/static/site-assets/downloads/cryptocurrency-whitepaper.pdf.
12. Litinski, "How to compute a 256-bit elliptic curve private key with only 50 million Toffoli gates," https://arxiv.org/pdf/2306.08585.
13. Cain, Xu, King et al, "Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits," https://arxiv.org/pdf/2603.28627.
14. National Institute of Standards and Technology, "Transition to Post-Quantum Cryptography Standards,"
15. Google Blog, "Quantum frontiers may be closer than they appear," https://blog.google/innovation-and-ai/technology/safety-security/
16. Cloudflare Blog, "Cloudflare targets 2029 for full post-quantum security," https://blog.cloudflare.com/post-quantum-roadmap/.
17. National Security Agency, "Announcing the Commercial National Security Algorithm Suite 2.0," https://media.defense.gov/2025/May/30/2003728741/-1/-1/0/CSA_CNSA_2.0_ALGORITHMS.PDF.
18. National Cyber Security Centre, "Timelines for migration to post-quantum cryptography," https://www.ncsc.gov.uk/guidance/pqc-migration-timelines.
19. NIS Cooperation Group, "A Coordinated Implementation Roadmap for the Transition to Post-Quantum Cryptography," https://digital-strategy.ec.europa.eu/en/library/coordinated-implementation-roadmap-transition-post-quantum-cryptography.
20. Bundesamt für Sicherheit in der Informationstechnik, "BSI empfiehlt Ende klassischer asymmetrischer Verschlüsselungsverfahren," https://www.bsi.bund.de/DE/Service-Navi/Presse/Pressemitteilungen/Presse2026/260211_Ende_klassischer_Verschluesselungsverfahren.html.
21. Canadian Centre for Cyber Security, "Roadmap for the migration to post-quantum cryptography for the Government of Canada." https://www.cyber.gc.ca/en/guidance/roadmap-migration-post-quantum-cryptography-government-canada-itsm40001.
22. Australian Signals Directorate, "Planning for post-quantum cryptography," https://www.cyber.gov.au/business-government/secure-design/quantum/planning-for-post-quantum-cryptography.
23. National Cybersecurity Office, "Cybersecurity Strategy issued in December 2025," https://www.cas.go.jp/jp/seisaku/pqc/pdf/report_202511.pdf.
24. Microsoft, "Quantum-safe security: Progress towards next-generation cryptography," https://www.microsoft.com/en-us/security/blog/2025/08/20/quantum-safe-security-progress-towards-next-generation-cryptography/.
25. Project Eleven, https://www.projecteleven.com/bitcoin-risq-list.
26. Ledger, "How Many Bitcoin Are Lost," https://www.ledger.com/academy/topics/economics-and-regulation/how-many-bitcoin-are-lost-ledger.
27. GitHub Bitcoin Improvement Proposals repository, https://github.com/bitcoin/bips.
28. Nick, "SHRIMPS: 2.5 KB post-quantum signatures across multiple stateful devices," https://blog.blockstream.com/shrimps-2-5-kb-post-quantum-signatures-across-multiple-stateful-devices/.
29. Vitalik Buterin, @vitalik.eth on X, https://x.com/VitalikButerin/status/2027075026378543132.
30. Ethereum Foundation, "Ethereum Strawmap," https://strawmap.org.
31. blueshift, GitHub, https://github.com/blueshift-gg/solana-winternitz-vault.
32. Alogrand Algokit Lora, https://lora.algokit.io/mainnet/transaction/7LLFUXNQUPYVWLJK3AVFPUM3BBTF3IQ2URIHHXYM7KOZNSL2CDXQ.

Investing in digital assets involves significant risks due to their extreme price volatility and the potential for loss, theft, or compromise of private keys. The value of the investment is closely tied to acceptance, industry developments, and governance changes, making them susceptible to market sentiment. A disruption of the internet or a digital asset network would affect the ability to transfer digital assets, and, consequently, would impact their value.

This material represents an assessment of the market environment as of the date indicated; is subject to change; and is not intended to be a forecast of future events or a guarantee of future results. This material is not intended to be relied upon as a forecast, research or investment advice, and is not a recommendation, offer or solicitation to buy or sell any securities or to adopt any investment strategy. The opinions expressed are as of a certain date and may change as subsequent conditions vary. The information and opinions contained in this material are derived from proprietary and nonproprietary sources deemed by BlackRock to be reliable, are not necessarily all-inclusive and are not guaranteed as to accuracy. As such, no warranty of accuracy or reliability is given and no responsibility arising in any other way for errors and omissions (including responsibility to any person by reason of negligence) is accepted by BlackRock, its officers, employees or agents. This material may contain "forward-looking" information that is not purely historical in nature. Such information may include, among other things, projections and forecasts. There is no guarantee that any of these views will come to pass. Reliance upon information in this material is at the sole discretion of the viewer.

The information provided is not intended to be tax advice. Investors should be urged to consult their tax professionals or financial professionals for more information regarding their specific tax situations.

©2026 BlackRock, Inc. or its affiliates. All Rights Reserved. **BLACKROCK** is a trademark of BlackRock, Inc. or its affiliates. All other trademarks are those of their respective owners.

BlackRock®